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3.R.R1 (1/1 point)

What is the difference between $\text{lm}(y \sim x*z)$ and $\text{lm}(y \sim \text{l}(x*z))$, when x and z are both numeric variables?

- ☐ The first one includes an interaction term between x and z , whereas the second uses the product of x and z as a predictor in the model. ✓
- ☐ The second one includes an interaction term between x and z , whereas the first uses the product of x and z as a predictor in the model.
- ☐ The first includes only an interaction term for x and z , while the second includes both interaction effects and main effects.
- ☒ The second includes only an interaction term for x and z , while the first includes both interaction effects and main effects. ✓

EXPLANATION

An interaction term between a numeric x and z is just the product of x and z . The difference is that in the first model, lm processes the "*" operator between variables and automatically includes main effects, whereas in the latter model, the expression inside the $\text{l}()$ function is not parsed as a part of the formula, but rather is simply evaluated.

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